



Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU

North Avenue, Diliman, Quezon City, Philippines
Tel. No. (+63 2) 8928-8642/ 8928-8937 Fax. No. (+63 2) 8920-1635 E-mail: central@mgb.gov.ph

LANDSLIDE AND FLOOD THREAT ADVISORIES

To: **Brgy. Chairperson**
Brgy. Jesus dela Peña
City of Marikina
Metropolitan Manila

Date: Oct 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **landslide and flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Malvar	14°37'50.4" N 121°05' 5.9" E	Very High	Brgy. Jesus dela Peña is situated along the Marikina River with the south portion within an innerbend of the river. This makes the area highly susceptible to flooding.
Austin	14°37' 51.2" N 121°05' 6.8" E	Very High	General Malvar area was submerged in 4 m floodwaters during typhoon Ondoy. The same area was inundated up to 3 m during typhoon Ulysses. Floodwaters typically recede after 12 hours. Source of the floodwaters is the covered creek running adjacent to the houses.
Riverside	14°38' 2.2" N 121°05' 5.3" E	Very High	Brgy. Jesus dela Peña is situated to the north of Brgy. Tañong. A portion of the dike failed during typhoon Ulysses which also affected areas in Brgy Jesus dela Peña. These are mainly along the boundary between the two barangays and along the riverside.
Remarks / Recommendations: ✓ Implement pre-emptive evacuation by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons. ✓ Construction and/or improvement of drainage system. Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal			

on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.

- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
- ✓ **Construction of appropriate and well-studied engineering measures** (e.g. dike or flood wall etc.) should be implemented to prevent overtopping of floodwaters and the fast rate of erosion along rivers/creeks. Make sure that the structure follows engineering standards and has structural integrity. However, even though there are engineering measures, communities near the river/creeks must remain vigilant and not complacent during strong typhoons and inclement weather.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.
- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.

Prepared by:


JULIUS VINCENT UMALI
Senior Geologist

MA. LOURDES STEPHANIE V. LEIDO
Senior Geologist

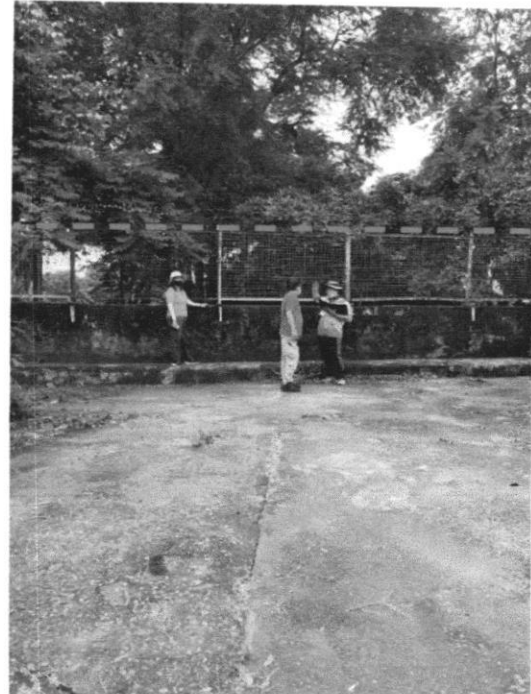

JETHRO B. CAPINO
Geologist II

Received by:

Designation:

Office:

PHOTOS



Inundated area in General Malvar. Floodwaters reached the first floor during typhoon Ulysses (above). Flood Height during typhoon Ulysses near the segment of the dike that collapsed during typhoon Ulysses (right)

Recommendations:

- ✓ **Implement pre-emptive evacuation** by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRMO) during monsoon (heavy rainfall) and typhoon seasons.
- ✓ **Construction and/or improvement of drainage system.** Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.
- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRMO, Office of the Civil Defense, and other related agencies.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.

- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.